IAM-09: Troubleshooting Access Issues

> **Series:** IAM — IAM Administration | **Notebook:** 9 of 12 | **Created:** January 2026 | **Last Updated:** 10/14/2026

Systematic Diagnosis of IAM Problems

Access issues are among the most common support requests. This notebook provides a systematic methodology for diagnosing and resolving IAM-related problems including permission denials, policy conflicts, and boundary issues.

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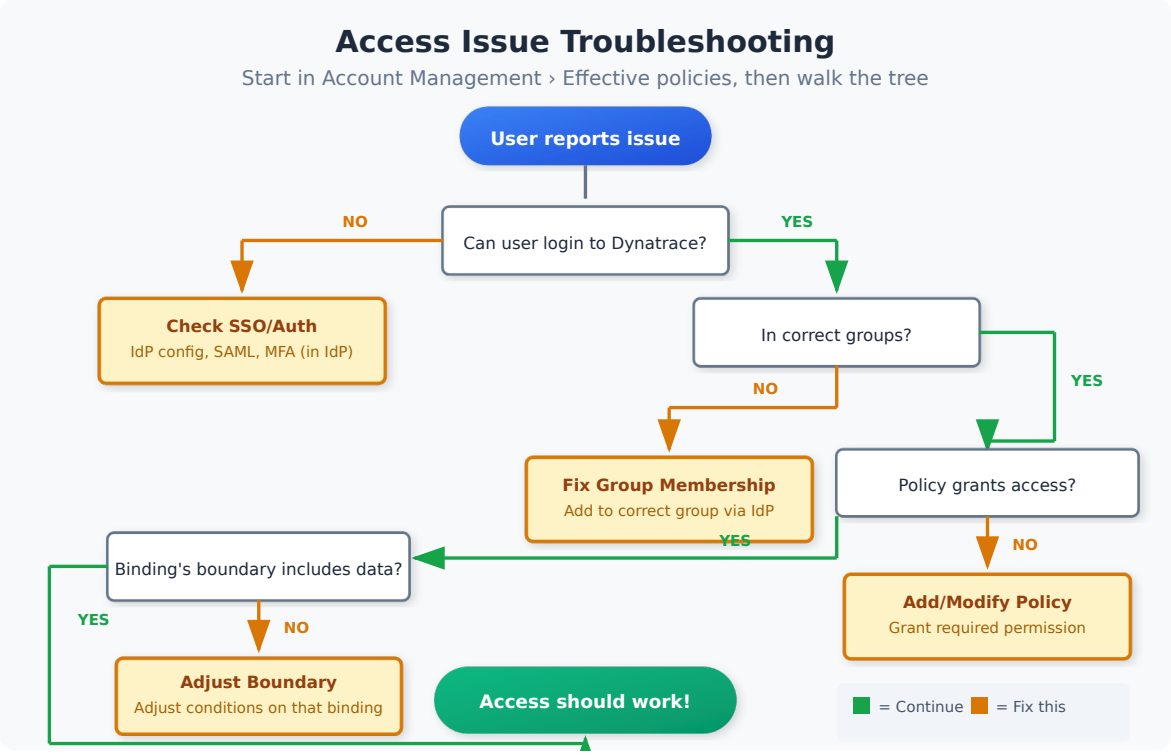
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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM
Permissions	`account-user-management` for full diagnosis
Knowledge	IAM-01 through 08 completed

1. Troubleshooting Methodology

Follow a systematic approach to diagnose access issues.



Step-by-Step Methodology

Step	Check	Tools
0	See which permissions the user actually holds	Account Management → Identity & access management → **Effective policies**
1	User can authenticate	SSO logs, IdP logs

Step	Check	Tools
2	User exists in Dynatrace	Account Management → Users
3	User is in correct groups	Account Management → Groups
4	Groups have correct policies	Account Management → Policies
5	Policies grant required permissions	Effective policies
6	Boundaries on the granting bindings don't exclude the data	Boundary configuration on each binding
7	No conflicting policies	Policy evaluation
8	Data exists and is accessible	DQL query

Start with Effective policies. Account Management lists it under Identity & access management as the place to *learn which permissions are assigned to users through policies and groups*. It answers steps 3–5 in one view before you open individual groups and policies.

> **Sources:** Manage users, groups, and permissions (DT docs) — *"Effective policies" to learn which permissions are assigned to users through policies and groups*.

Information Gathering

Before troubleshooting, collect:

```
User Information:
☐ User's email address
☐ What action they're trying to perform
☐ Exact error message received
☐ When the issue started
☐ What changed recently (role, team, etc.)

Environment Information:
☐ Which environment(s) affected
☐ What data/resource they're accessing
☐ Screenshot of error (if available)
```

2. Common Access Issues

Issue Categories

Category	Symptoms	Common Cause
Authentication	Can't login	SSO misconfigured, account disabled
Authorization	Access denied	Missing policy, wrong group
Data Access	No results	Boundary restriction, no data
Partial Access	See some, not all	Boundary or policy condition
Intermittent	Sometimes works	Caching, eventual consistency

Quick Diagnosis Table

Symptom	Likely Cause	First Check
"Access Denied"	Missing permission	Policy assignments
"Forbidden"	Boundary restriction	Boundary config
"Not Found"	No data or no access	Query without filters
"Unauthorized"	Token/auth issue	Token validity
Can't see entity	Boundary or policy condition	Entity's `dt.security_context` vs. the boundary
Can read, can't write	Read-only policy	Policy permissions

3. Permission Denied Debugging

When users receive "Access Denied" or "Forbidden" errors.

Diagnostic Steps

Step 1: Verify User Exists

1. Go to Account Management → Users
2. Search for user by email
3. Verify account is active (not disabled)

Step 2: Check Group Membership

1. Click on user to view details
2. Note all groups user belongs to
3. Verify expected groups are present

Step 3: Review Group Policies

1. For each group, check assigned policies
2. Verify policies grant needed permissions
3. Check policy conditions/restrictions

Step 4: Check Boundaries

1. Review boundaries assigned to groups
2. Verify boundaries include target resources
3. Check boundary conditions

Permission Evaluation Order

```
1. User authenticated? → No = Auth failure
2. User in any groups? → No = No permissions
3. Any DENY statement matches? → Yes = Access denied (boundaries never apply
to DENY)
4. Any policy grants permission? → No = Access denied
5. Boundary on the granting binding excludes the data? → Yes = Check boundary
scope
6. Access granted
```

Common Permission Issues

Issue	Symptom	Resolution
Not in group	No access at all	Add user to group via IdP
Wrong group	Partial access	Move to correct group
Policy too restrictive	Can't perform action	Modify or add policy

Issue	Symptom	Resolution
Missing environment	Can't access env	Add env to policy scope

4. Policy Conflict Resolution

When multiple policies interact unexpectedly.

How Policy Evaluation Works

1. Collect all policies from all user's groups
2. Evaluate each policy's conditions
3. **Any ALLOW** = access granted (unless explicitly denied)
4. **No matching ALLOW** = access denied

Conflict Scenarios

Scenario	Policy A	Policy B	Result
Both allow	ALLOW read	ALLOW write	ALLOW both
One allows	ALLOW read	No match	ALLOW read
Overlapping	ALLOW all envs	ALLOW prod only	ALLOW all envs
Condition mismatch	ALLOW (condition: A)	ALLOW (condition: B)	Depends on conditions

Debugging Policy Conflicts

Step 1: List All Applicable Policies

1. Identify all groups user belongs to
2. List all policies assigned to those groups
3. Note overlapping permissions

Step 2: Evaluate Each Policy

For each policy, check:
- Does it apply to the target resource?

- Does it grant the required permission?
- Are conditions met?

Step 3: Identify Conflict

Conflict Type	Indicator	Resolution
Unintended restriction	User in wrong group	Remove from group
Missing permission	No policy grants access	Add policy
Condition not met	Policy exists but doesn't apply	Adjust condition
Scope mismatch	Policy applies to wrong scope	Correct scope

5. Boundary Troubleshooting

Boundaries restrict what data users can access within their permissions. A boundary is attached to a **policy binding** — the assignment of a policy to a group — not to a user or a group as a whole. The same user can therefore hold one permission through a bounded binding and the same permission through an unbounded one.

Boundary Evaluation

```
User wants to access data
  ↓
A DENY statement matches? → Yes = Access denied (boundaries don't apply to
DENY)
  ↓ No
Any binding grants the permission? → No = Access denied
  ↓ Yes
Is any granting binding unbounded? → Yes = That binding's policy applies
without boundary conditions
  ↓ No (every granting binding has a boundary)
Does the boundary have conditions for this permission's service? → No =
Unconditional for that binding
  ↓ Yes
Data within the boundary conditions? → No = Data hidden
  ↓ Yes
Data returned
```

Two consequences catch most teams out. A service (domain) left out of a boundary is **unscoped, not blocked**: only conditions added for a service are applied to that service's

permissions. And when more than one boundary applies to a policy, effective statements are calculated for each boundary separately, so two narrow boundaries on one binding can each leave the other's domain unconditional.

> **Sources:** IAM policy boundaries (DT docs) — *"Boundaries don't apply to DENY statements."*; *"Only conditions added to service configuration will be applied to relative permissions."*; *"When more than one boundary applies to a policy, then effective statements are calculated for each boundary separately."*

Common Boundary Issues

Issue	Symptom	Cause	Resolution
Too restrictive	See no data	Boundary too narrow	Expand boundary
Not restrictive enough	See too much	Boundary too broad, a service missing from the boundary (unscoped), or another binding grants the same permission without a boundary	Narrow the boundary, add conditions for every service the policy uses, and check Effective policies for unbounded bindings
Segment applied	Some data missing	A segment the user selected hides data they are allowed to see	Clear the segment selector before debugging access
Environment excluded	No env access	Env not in boundary	Add environment

Boundary Debugging Steps

Step 1: Identify Boundary

1. Find user's groups
2. Check the boundary attached to each policy binding of those groups (boundaries are per binding)
3. Note boundary scope (environment, storage, settings)

Step 2: Verify Boundary Scope

For each domain:
- **Environment**: Which environments included?
- **Storage**: Which data types included?
- **Settings**: Which settings accessible?

Step 3: Test Without Boundary

Temporarily:
1. Add user to unbounded admin group
2. Test if access works
3. If yes, boundary is the issue
4. Remove from admin group

Step 4: Adjust Boundary

Modify boundary to include:
- Missing environments
- Missing security-context values
- Missing data types

6. SSO and Authentication Issues

When users can't login at all.

SSO Troubleshooting

Symptom	Check	Resolution
Redirect loop	IdP configuration	Verify ACS URL
SAML error	SAML response	Check attribute mapping
Signs in but has no access	Group claims vs. each group's SAML Group Attribute Value	Users are created automatically at SAML sign-in; fix the group mapping so a claim matches
MFA failure	Your IdP's MFA policy	Dynatrace does not configure MFA; fix it in the IdP
Session expired	Session settings	Extend session duration

SAML Debugging

Check SAML Response:

1. Use browser dev tools → Network
2. Find SAML response during login
3. Decode and verify:
   - Subject/NameID (email)
   - Groups attribute
   - Conditions (timestamps)

Common SAML Issues:

Issue	Indicator	Fix
Wrong NameID	Email doesn't match	Configure NameID format
Missing groups	Groups not in assertion	Add groups attribute in IdP
Expired assertion	Time mismatch	Sync clocks, extend validity
Invalid signature	Signature error	Update IdP certificate

Group Mapping Issues

If user logs in but has wrong permissions:

1. Check SAML groups in assertion
2. Verify group mapping configuration
3. Confirm IdP group names match exactly
4. Check for case sensitivity

7. Token and API Access Issues

When API tokens or OAuth clients don't work as expected.

Token Issues

Error	Cause	Resolution
`401 Unauthorized`	Token invalid/expired	Regenerate token
`403 Forbidden`	Missing scope	Add required scope
`404 Not Found`	Wrong endpoint or no data	Verify API path
Token rejected	Token revoked	Create new token

Token Debugging Steps

Step 1: Verify Token Valid

1. Check token hasn't expired
2. Verify token not revoked
3. Confirm token is for correct environment

Step 2: Check Scopes

1. List token's scopes
2. Compare to required scopes for API
3. Add missing scopes if needed

Step 3: Test with curl

```
# Connectivity smoke test — pick an endpoint that survives the Gen3 upgrade
curl -H "Authorization: Api-Token &lt;TOKEN&gt;" \
  https://&lt;env&gt;.live.dynatrace.com/api/v2/settings/schemas
```

> **Do not use `/api/v2/entities` for this test.** It is flagged blocked at upgrade, so on a latest-Dynatrace tenant it fails for a reason that has nothing to do with the token — sending you to debug scopes and expiry when the endpoint itself is gone. `/api/v2/settings/schemas` carries forward and needs only `settings.read`, which makes it a cleaner probe. To check an entity actually resolves, use DQL (`smartscapeNodes`) rather than the classic Entities API — see FAQ entry 16.

OAuth Client Issues

Issue	Check	Resolution
Can't get token	Client credentials	Verify client ID/secret
Wrong permissions	Client scopes	Add required scopes
Token expired	OAuth token expiry is not configurable	Request a new token from the client

8. Diagnostic Queries

DQL queries to help diagnose IAM issues.

```
// Find recent access denied events
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-24h
| filter event.kind == "AUDIT_EVENT"
| filter event.outcome == "403"
| fields timestamp, user.id, resource, event.type, origin.address
| sort timestamp desc
| limit 25
```

```
// Recent authenticated API calls (credential type per call)
// Retitled 10/12/2026: authentication.type is set on every audited request,
so this lists API traffic.
// To list sign-ins only, filter event.type == "LOGIN".
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter isNotNull(authentication.type)
| fields timestamp, user.id, authentication.type, authentication.grant.type,
event.outcome
| sort timestamp desc
| limit 50
```

```
// Failed sign-in attempts (event.type LOGIN with outcome "failure")
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter event.type == "LOGIN" and event.outcome == "failure"
| fields timestamp, user.id, authentication.type, origin.address
| sort timestamp desc
| limit 50
```

```
// Recent IAM API activity in this environment (account-level policy or group
changes: account audit log)
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter in(event.type, {"POST", "PUT", "PATCH", "DELETE", "CREATE",
"UPDATE"}) and contains(resource, "iam")
| filter not startsWith(resource, "/lookups/")
| fields timestamp, user.id, event.type, resource
| sort timestamp desc
| limit 25
```

```
// Token lifecycle: API and ActiveGate token creation, update and deletion
// Corrected 10/13/2026: token lifecycle events carry event.provider API_TOKEN
/ ACTIVE_GATE_TOKEN,
// and their resource is the token ID. The previous filter
(authentication.type == "TOKEN" or
// contains(resource, "token")) matched every API call made with a token
instead, plus scheduled
// /lookups/iam_tokens uploads. Same filter as IAM-06 and IAM-07.
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter in(event.provider, {"API_TOKEN", "ACTIVE_GATE_TOKEN"})
| fields timestamp, user.id, event.provider, event.type, resource,
event.outcome
| sort timestamp desc
| limit 25
```

```
// All audit events in the last hour (immediate troubleshooting)
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-1h
| filter event.kind == "AUDIT_EVENT"
| fields timestamp, user.id, event.type, resource, event.outcome
| sort timestamp desc
| limit 50
```

Troubleshooting Checklist

Use this checklist when debugging access issues:

```
AUTHENTICATION
☐ User can access IdP login page
☐ User can authenticate with IdP
☐ SAML assertion is valid
☐ User account exists in Dynatrace
☐ User account is active (not disabled)

GROUP MEMBERSHIP
☐ User is in expected groups
☐ Groups are synced from IdP (if using SCIM/JIT)
☐ No unexpected group memberships

POLICIES
☐ Groups have policies assigned
☐ Policies grant required permissions
☐ Policy conditions are met
☐ No conflicting policies

BOUNDARIES
☐ Boundary includes target environment
☐ Boundary includes target data type
☐ No overly restrictive boundaries

DATA
☐ Data exists in target timeframe
☐ No segment is selected that hides the data (segments filter views, they do
not restrict access)
☐ Query syntax is correct
```

Escalation Path

When self-service troubleshooting doesn't resolve the issue:

Level	Who	When
L1	Team lead	Can't determine cause
L2	IAM admin	Need policy/boundary changes
L3	Platform team	Infrastructure/IdP issues
L4	Dynatrace support	Product issues

Information for Escalation

Include in escalation:

```
☐ User email/identifier
☐ Exact error message
☐ Steps to reproduce
☐ Timestamp of issue
☐ What was already checked
☐ Audit log excerpts
☐ Screenshots
```

Series Conclusion

Congratulations! You've completed the IAM series.

Series Recap

Notebook	Key Topics
01: Governance Foundations	Framework, roles, models
02: SSO and Authentication	SAML, IdP integration, MFA
03: Group Architecture	Hierarchy, naming, SAML mapping
04: Policy Authoring	Syntax, conditions, GitOps
05: Boundary Design	Three domains, isolation patterns
06: User Lifecycle	SCIM, JIT, tokens
07: Audit and Compliance	Audit queries, SOC2/SOX/HIPAA
08: Multi-Environment	Account vs env, break-glass
09: Troubleshooting	Methodology, common issues
10: Templated Policy-Group Assignments	Parameterized policies, IAM API bindings, Monaco
11: Policy Persona Workshop	Persona-based policy design, schema audit, domain mapping

Notebook	Key Topics
12: API Provisioning & Validation	Provisioning scripts, binding reports, cleanup, validation DQL
95: Terraform IAM Provisioning (LAB)	Groups, policies, boundaries and bindings as Terraform
96: Python IAM Provisioning (LAB)	Step-by-step Account Management API provisioning in Python
99: Best Practice Summary	74 best practices with priorities and settings

Ongoing Maintenance

Task	Frequency
Access reviews	Quarterly
Policy review	Semi-annually
Boundary review	Semi-annually
Token rotation	Per policy (90 days typical)
Audit log review	Monthly
Documentation update	As needed

Summary

In this notebook, you learned:

- Systematic troubleshooting methodology
- Common access issue categories
- Permission denied debugging steps
- Policy conflict identification and resolution
- Boundary troubleshooting techniques
- SSO and authentication debugging
- Token and API access troubleshooting

- Diagnostic DQL queries

References

- IAM Troubleshooting
- SAML Configuration
- API Authentication
- IAM policy boundaries (DT docs)
- Token concepts (DT docs) — the IAM token types table lists *"Expiry configurable?"* as *"No"* for OAuth tokens

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